Forecasting Firm-Level Credit Demand in Germany

Predicting next-year growth in non-current liabilities using firm fundamentals, macro indicators, and text-based sentiment

Deep Dive | Machine Learning for Economists, Summer 2026 | [Anastasiia Hinzpeter, Thanh Huyen Dao, Kha Nhi Pham]

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EXECUTIVE SUMMARY

Firm fundamentals provide useful, but limited, signals of future financing changes

MAIN FINDINGS:

~5% Reduction in prediction error

Best-performing model vs. historical-mean benchmark

Existing debt structure drives the forecast

Long-term debt share, firm size and asset mix matter most. "Last year's growth continues" doubles the error.

Firm data dominate

Survey expectations offer limited and inconsistent gains; text sentiment adds little value

Uneven accuracy

Forecasting performs better for large firms than for smaller businesses

Implication for NRW.BANK

Financial-statement-based forecasting could support further development of financing-demand intelligence, but current performance, particularly for smaller firms, is insufficient for reliable standalone decisions.

METHODOLOGY & DATA

How we assessed the predictability of corporate financing changes

BUSINESS NEED **Early read on credit demand to plan financing programmes.**
Demand is unobservable, so the forecast target is one-year-ahead growth in non-current liabilities.

3 INFORMATION INPUTS | ADDED STEP BY STEP

Firm fundamentals
Annual firm's balance sheet from Amadeus/BvD

Survey expectations
Annualised economic expectations from ifo and EC surveys.

Text sentiment
Text-based sentiment and uncertainty from Bundesbank Monthly Reports

BINDING CONSTRAINT

1.74M firm-years

Macro and text indicators are constant within each calendar year.

9 development years **3** test years

Survey block is highly redundant
- 20 series → First 3 PCs explain 91.6%

RQ1 How responsive is credit demand to economic expectations, and can public-report sentiment improve forecasts?

RQ2 Does predictive value vary with firm characteristics such as size, leverage, and sector?

TEST DESIGN

Learn & tune: 2010–2019	Test once: 2020–22

Model frozen before the test; COVID falls entirely in the test years.

4 models
- Regularized linear regression
- Random Forest
- XGBoost
- Neural Network

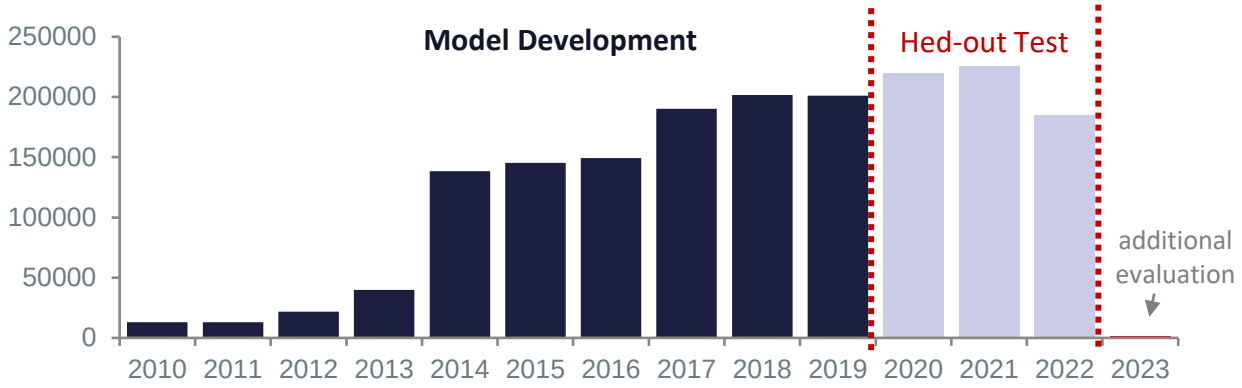
Validation
Time-ordered model development and evaluation on previously unseen fiscal years

METHODOLOGY & DATA

1.74 million firm-year observations enable time-ordered model evaluation with minimal data loss

DATA OVERVIEW

Development panel: firm-years per fiscal year



1,113,369
development firm-years

630,228
held-out firm-years

285,230
Unique firms

86.7%
of held-out firms already observed during development

Unbalanced firm panel with expanding coverage over time; reporting availability varies across years.
2023: Limited outcome coverage - excluded from headline results

DATA PIPELINE - FIRM-YEAR LEVEL

01 | OUR FIRM-LEVEL PROCESSING

99.97%
RETAINED

SAMPLE INTEGRITY
Removed only rows with non-positive total assets or a material balance-sheet identity violation. 484 / 1,75m removed (0.028%).

ECONOMIC FEATURES
Created financial ratios, firm-size indicators and historical dynamics. Changes calculated only for consecutive years.

02 | LEAKAGE-SAFE PREPROCESSING

Invalid values are not turned into artificial extremes

- Invalid ratios remain missing; missing predictors are handled by each model.
- Extreme values treated using model-specific methods.
- Winsorization, imputation, scaling and encoding fitted on training data only.

METHOD DISTINCTION

Invalid ratio → missing
Valid but extreme ratio → model-specific treatment

METHODOLOGY & DATA

Three information sets and out-of-time validation allow us to assess their predictive value

SURVEY AND TEXT DATA BECOME 26 ANNUAL EXPECTATION AND SENTIMENT INDICATORS

MERGED BY YEAR ONTO THE FIRM PANEL

Same values for every firm in a year. December of year t feeds the forecast for t+1, so there is no look-ahead.

Block 1: Firm fundamentals — 22 features → **Block 2: + Survey** — 42 features → **Block 3: + Text** — 48 features

Δ1: incremental survey value

Δ2: incremental text value

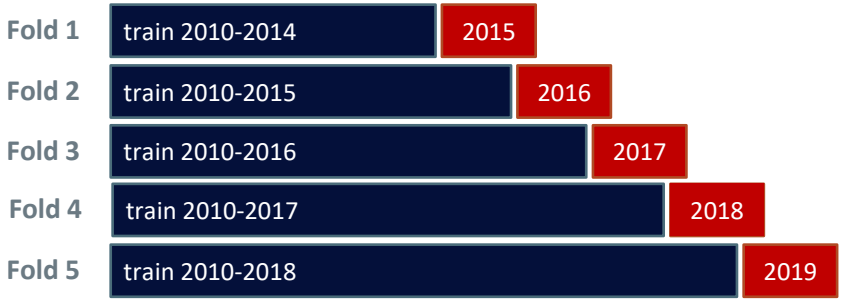
RQ1 is evaluated by measuring how forecast accuracy changes when moving from one block to the next

Firm → +Survey: value of macro expectations

+Survey → +Text: additional value of public-report sentiment

TIME-AWARE VALIDATION PREVENTS LOOK-AHEAD BIAS

Expanding-window CV for model development; preprocessing fitted within each training fold



Validation
→ tune hyperparameters
→ compare models
→ compare feature blocks
→ freeze specification

2020–2022 evaluated once, after the freeze

Test
→ estimate final OOS performance

2023 reported separately due to thin coverage

Evaluation metrics: Out-of-sample R^2 RMSE MAE

MODELING

Four model families span increasing flexibility

MORE STRUCTURED / TRANSPARENT — MORE FLEXIBLE

Penalized Linear	Random Forest	XGBoost	Neural Network
Linear + shrinkage	Bagged trees	Boosted trees sequentially	Learned nonlinear transformations

PENALIZED LINEAR

Transparent regularized benchmark

- Ridge, Lasso, Elastic Net
- Additive linear effects
- Shrinkage stabilizes correlated features

SELECTED · LASSO (1-SE RULE)

α = 0.0316 | 3/117 non-zero
Convergence verified

RANDOM FOREST

Nonlinear benchmark through bagging

- Many decorrelated regression trees
- Captures nonlinearities and interactions
- Reduces variance vs. a single tree

XGBOOST

Flexible boosted-tree model

- Trees fitted sequentially to remaining errors
- Captures nonlinearities and interactions
- Regularization controls tree complexity

NEURAL NETWORK

Flexible nonlinear benchmark

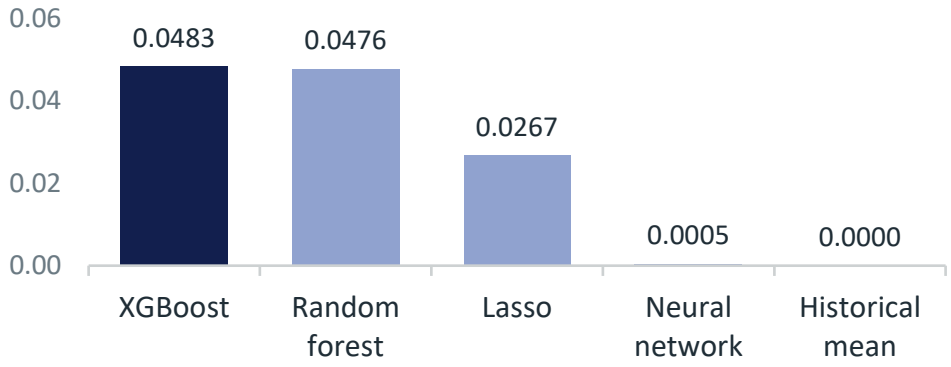
- Feed-forward network
- Learns nonlinear transformations and interactions
- Requires scaling and stronger regularization

All four families use the same outcome, nested information blocks and out-of-time evaluation framework; preprocessing and tuning are adapted to each algorithm.

OVERALL MODEL PERFORMANCE

Machine learning improves forecasts, but predictive accuracy remains limited

OUT-OF-SAMPLE R^2, HELD-OUT 2020–22 (ALL INFORMATION BLOCKS)



Model	RMSE ↓	MAE ↓	OOS R^2 ↑
Mean benchmark	0.3779	0.2779	0.0000
Lasso Regression	0.3728	0.2747	0.0267
Random Forest	0.3688	0.2771	0.0476
Neural Network	0.3778	0.2825	0.0005
XGBoost	**0.3686**	**0.2771**	**0.0483**

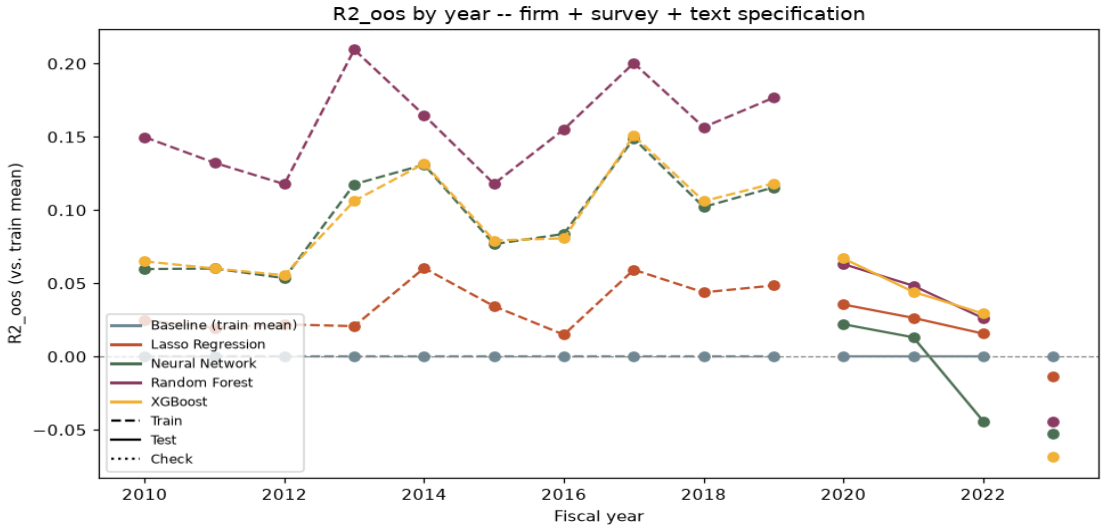
0.0483

XGBoost R^2 oos: 4.8% less squared error than the benchmark

0.3686 vs. 0.3779

RMSE of XGBoost vs. historical-mean benchmark (−2.5%)

R2 BY YEAR · HIGHER IS BETTER



WHAT THIS MEANS

Tree-based models (XGBoost, random forest) clearly beat the benchmark and the linear model, but most year-to-year variation in firms' financing remains unpredictable from annual accounts.

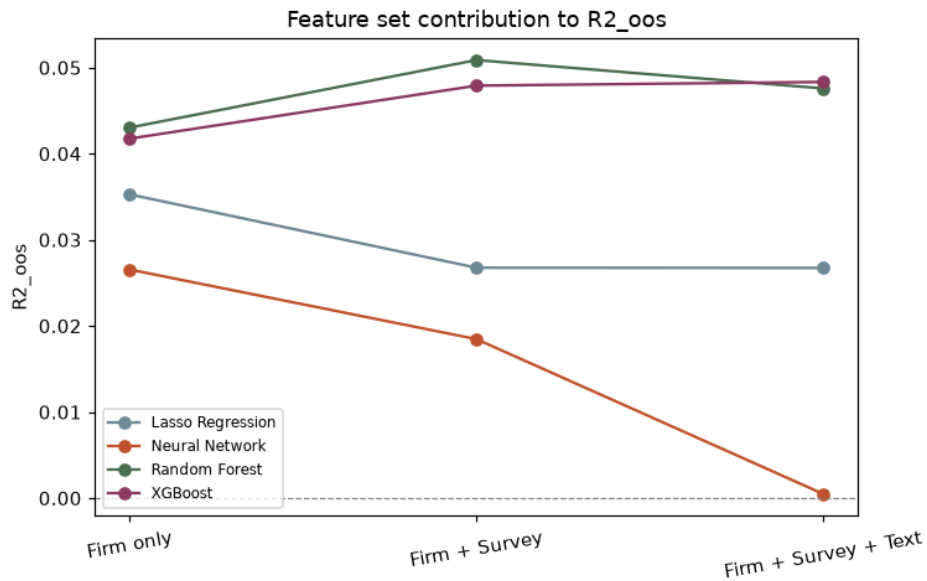
WHY THE IMPROVEMENT IS MODEST

Noisy target: liability growth is only a proxy for demand.

Regime shift: test years include COVID, unseen in training.

RQ1 - Does aggregate information improve credit-demand forecasts?

HELD-OUT R2 - HIGHER IS BETTER

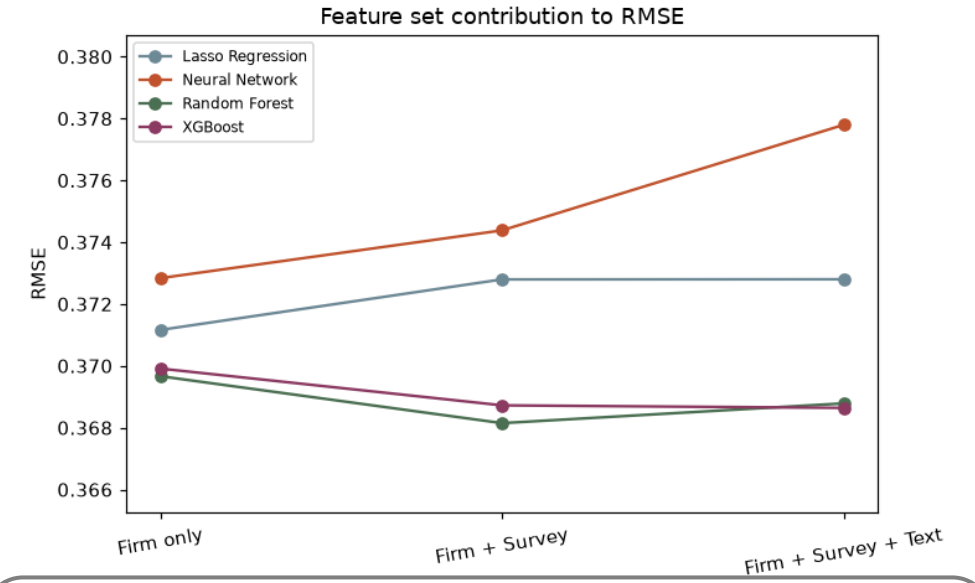


Δsurvey = RMSE(Firm) − RMSE(Firm + Survey)

Δtext = RMSE(Firm + Survey) − RMSE(Firm + Survey + Text)

Positive Δ means the added information improves prediction.

HELD-OUT RMSE – LOWER IS BETTER



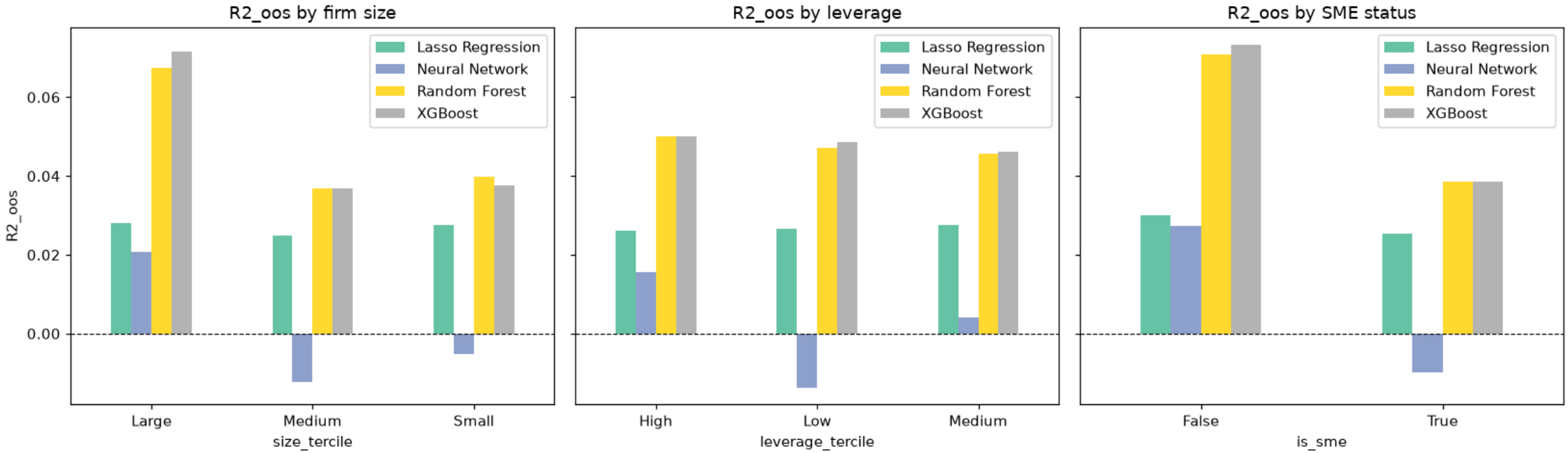
Model	Survey gain - RMSE	Text gain - RMSE
Lasso	-0.001634	-0.000004
Random Forest	0.001517	-0.000639
XGBoost	**+0.001189**	**+0.000083**
Neural Network	-0.001545	-0.003413

MAIN FINDINGS:

- **Different models learn different patterns**
- **XGBoost can relatively gain information from macro data and text data. The reason may be because XGBoost uses native missing-value handling and unscaled engineered features. Another reason can be XGBoost model is tuned separately across feature sets**

RQ2 - Does the predictive value vary over firm characteristics?

MODEL PERFORMANCE BY FIRM CHARACTERISTICS



DIMENSION DEFINITION

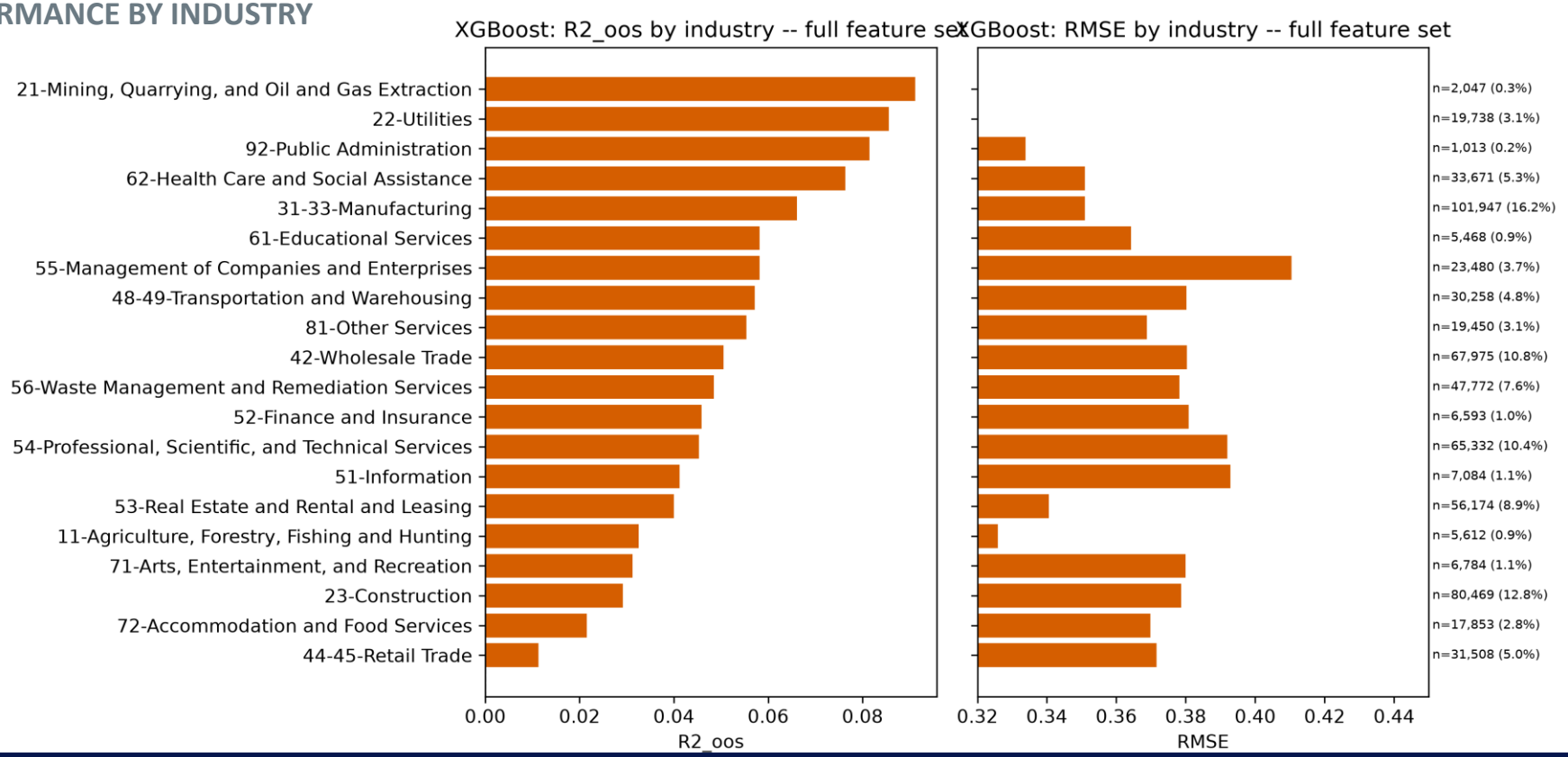
- **Firm size:** terciles by total assets at 33% and 66% percentile
- **Leverage:** terciles by (non-current + current liabilities) / total assets
- **SME:** total assets ≤ €43M (by NRW.BANK's definition)

MAIN FINDINGS

- **All models perform best with large firms**
- **LASSO have less R2 heterogeneity**
- **Unlike firm size, capital structure (leverage) creates less variation in predictive accuracy**

RQ2 - Does the predictive value vary over firm characteristics?

MODEL PERFORMANCE BY INDUSTRY

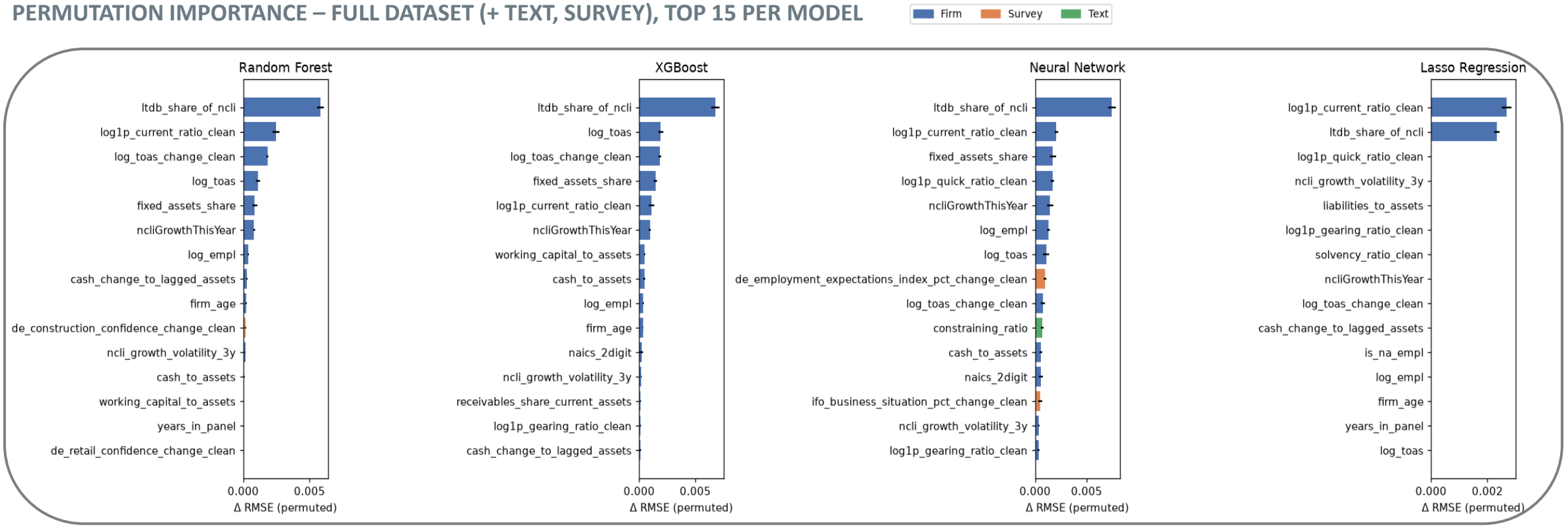


MAIN FINDINGS:

- **The top ranking** (Mining 0.3%, Public Administration 0.2%, Utilities 3.1%) is **dominated by small-share industries.** The performance may be caused from noise data
- **R^2 and RMSE is not consistent in some cases**, because R^2oos measures improvement over train set's variance, not absolute accuracy.

Model interpretation: what actually drives the forecasts

PERMUTATION IMPORTANCE – FULL DATASET (+ TEXT, SURVEY), TOP 15 PER MODEL



MAIN FINDINGS:

- **The composition of existing debt dominates**
- Marginal impact of survey/text block: **Firm-level financial data heavily dominates** the top 15 features
- **Lasso's L1 penalty explicitly induces sparsity**
- **The Neural Network tends to spread weight across all of features.** It hasn't found one dominant, decisive relationship and is instead patching together many weak, partially redundant signals.

CONCLUSIONS & NEXT STEPS

Better demand measures and targeted validation could strengthen future forecasts

THREE PRIORITIES

1

Use more direct credit-demand data

Replace liability growth as a proxy with loan applications, approvals or drawdowns from NRW.BANK and partner banks.

2

Improve prediction for smaller firms

Add information annual accounts miss, such as intra-year financials or account flows, and test SME-specific models.

3

Test aggregate and NRW-relevant forecasts

Check how well forecasts add up to segment-level demand, and evaluate performance for NRW-based firms and bank clients.

KEY TAKEAWAY

Firm balance sheets already provide a measurable early signal of financing needs; better demand data and validation on NRW.BANK's own portfolio are what would turn it into a decision-support tool.

THANK YOU!